

VICKEY GHIMIRE

✉ jung.vickey@gmail.com

🌐 [linkedin.com/in/vickeyghimire](https://www.linkedin.com/in/vickeyghimire)

🐙 github.com/junggeyy

Education

M.S. in Geospatial Information Science

The University of Texas at Dallas

08/2026 - 05/2028

Richardson, TX

B.S. in Computer Science

Northern Kentucky University | GPA: 3.94/4.0

08/2022 - 05/2026

Highland Heights, KY

Technical Skills

Geospatial	ArcGIS Pro, ArcGIS, QGIS, GeoPandas, Rasterio, rioxarray, Xarray, Spatial SQL, Raster Analysis, DEM/Terrain Analysis, OSM, OSRM
Programming	Python, R, C++, Java, SQL, JavaScript
Data & ML	Pandas, NumPy, PyTorch, SciPy, scikit-learn, Time-Series Modeling, NLP
Web & Cloud	Django, FastAPI, PostgreSQL, REST APIs, AWS (ECS, RDS), Firebase

Experience

High Performance Computing Student Assistant

08/2026 – Present

High Performance Computing (HPC), UTD

Richardson, TX

- Provide technical support to researchers and students using HPC systems, troubleshooting Linux/UNIX environments, software compilation/installations, and scientific computing applications in coordination with HPC staff.
- Monitor HPC systems and usage data, assisting with system reporting, data visualization, and user support while contributing to technical workshops and research-computing training activities.

Full-Stack Django Developer

09/2025 – 05/2026

Department of Physics, NKU

Highland Heights, KY

- Developed a modular Django-based web application, refactoring a legacy script into an MTV-compliant architecture, improving code maintainability and enabling faster feature development.
- Built and integrated spatial data pipelines using OpenStreetMap and NASA POWER APIs to replace simulated inputs with site-level environmental data, improving accuracy of optimization workflows.
- Optimized a genetic-algorithm optimization workflow with convergence-based termination and result caching, cutting average computation time by **35%**.

Research Assistant- Data Analytics

10/2024 – 08/2026

Human-Computer Interaction Lab, NKU

Highland Heights, KY

- Designed an end-to-end novelty + relevance detection framework for LLM-generated content using hybrid semantic similarity (Jaccard + BERT), enabling automated content review of 9,000+ topics with **90%+** coverage.
- Led analysis of ChatGPT's memory feature across **135+** users, building reproducible pipelines for multi-account data cleaning, transformation, automated visualization, and response analysis.

Business Services Supervisor

08/2025 – 05/2026

NKU Campus Recreation Center

Highland Heights, KY

- Supervised and trained **25+** student employees across front-desk & membership services at NKU's largest recreation facility; supported customer inquiries, membership sales, phone communications, and day-to-day facility operations.

Projects

Site Suitability Analysis: Fire Station in Kathmandu | *GeoPandas, xarray, rioxarray, rasterio* | [Link] July 2026

- Built a multi-criteria site-suitability pipeline by aligning OSM vector data and SRTM DEM rasters, deriving slope and Euclidean-distance to quantify terrain safety and emergency accessibility, to support operation and faster response time.
- Applied Weighted Linear Combination to normalized suitability scores, ranking candidate fire-station locations and exporting results to an interactive Leaflet map for ground evaluation.

Disaster-Aware Road Risk Viewer | *GeoPandas, React-Leaflet, Shapely, FastAPI* | [Link]

March 2026

- Developed a geospatial decision-support platform visualizing seasonal landslide/flood risk in Nepal's Bagmati Province.
- Segmented OSM road networks and derived slope and Terrain Ruggedness Index features to score risk, then matched OSRM routes to pre-scored segments for comparative route-safety analysis for possible travel paths.

Predictive Maintenance Framework | *Python, Scikit-learn, Pandas, MCP servers* | [Link]

December 2025

- Built a two-stage predictive maintenance pipeline using Random Forest models to detect machine failures within 24-hours and estimate short-term Remaining Useful Life (RUL), achieving F1-score of **0.93** and R^2 of **0.98** for RUL prediction.
- Engineered temporal features from industrial telemetry and event logs (rolling stats, lag signals, maintenance histories) to capture pre-failure patterns that raw sensor data alone could not.